TELOMY

Engineering Guide

Platform · Web · Design System · Infrastructure

Version 3.0 · 29 September 2026 · Supersedes v2.0

What's new in v3.0

- Web engineering section (Next.js 15 + Sanity + Vercel + Cloudinary + Plausible)
- Design system integration layer (references UI Design System v1.0)
- Cross-surface auth and consent model (mobile app ↔ web ↔ Telomy Care)
- Frontend web engineer added to the 6-month build plan
- Updated open decisions for the incoming CTO
- Sinc's role clarified across services (drafts, never signs)

Companion documents

- Master DPR v5.0 — strategy, positioning, commercial model
- Brand Architecture v1.0 — product family and naming
- UI Design System v1.0 — visual + interaction system across surfaces
- Website Development Guide v1.0 — telomy.com specification
- Chamber Spec v4.2 — Halo hardware and firmware (unchanged, locked)

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1. Overview & scope

This guide is the engineering-side companion to the Telomy Master DPR v5.0. It supersedes Engineering Guide v2.0 and reflects the mature state of Telomy planning as of September 2026 — after the brand is locked, the website architecture is defined, and the design system has a canonical source.

In scope

- Cross-surface system architecture (mobile app + web + backend + Halo firmware)
- The event-first data model and multi-source ingest pipeline
- The AI stack: Sinc as named companion, N-of-1 study engine, pattern surfacing, longevity age
- Mobile app engineering (iOS + Android via React Native + Expo)
- NEW: Web engineering — telomy.com (marketing) and Telomy Care (clinician web)
- Backend services, storage, and APIs
- Design system integration (shared tokens + component behaviour across surfaces)
- Security, DPDP compliance, key management, audit
- DevOps: CI/CD, environments, monitoring, incident response
- Team, hiring, and updated 6-month build plan (now includes frontend web engineer)

Out of scope (covered elsewhere)

- Telomy Halo firmware / hardware (Chamber Spec v4.2)
- Telomy Formulas nutraceutical formulation (DPR §10)
- Clinical protocol design (DPR §8; CMO-owned)
- Brand and identity system (Brand Architecture + Brand Identity Brief)
- Visual design of screens and components (UI Design System v1.0)
- Business, legal, and financial planning (DPR)

2. System architecture

2.1 The five surfaces + one backbone

Telomy is one platform running across five user-facing surfaces, all backed by a single set of backend services. Every surface reads and writes to the same Vault, but presents it differently.

Surface	Tech	Primary user
Mobile app (iOS + Android)	React Native + Expo	The person whose biology is being tracked
telomy.com (marketing website)	Next.js 15 + Sanity CMS	Prospective users, press, investors, clinicians
Telomy Care (clinician web)	Next.js 15 (separate app)	Clinicians reviewing patient panels
Halo touchscreen (in-chamber)	React embedded on ESP32 web-view	User during Halo session
Future: Band / Patch / Ring firmware displays	Rust + LVGL	Wearer at-glance

2.2 Seven-layer backend stack (unchanged from v2.0)

Layer	Purpose	Primary tech
Client	Mobile app, web apps, embedded firmware	React Native, Next.js 15, ESP32 firmware
Edge	On-device inference, local Vault cache, voice STT	Core ML / TFLite, WhisperKit / whisper.cpp
Gateway	Auth, rate-limit, DPDP consent enforcement	AWS API Gateway or Cloudflare Workers
Services	Vault, Events, Study, Recommendation, Clinician, Sinc, Payments	FastAPI + Python 3.12 microservices
Event bus	Async ingestion, correlation triggers	AWS SQS + EventBridge (or NATS)
Data & ML	Time-series, documents, features, models	Postgres + TimescaleDB, S3, Feature store, MLflow
Integrations	Wearables, labs, Halo, payments, CMS	Terra API, Rook, Razorpay, Sanity webhooks

2.3 The three deployment domains

Domain	Serves	Deployed to
telomy.com	Marketing site, journal, manifesto, waitlist	Vercel (edge network with Mumbai POP priority)
app.telomy.com	Web-based product access (Telomy Care + web mobile fallback)	Vercel (same region, separate project)
api.telomy.com	Backend APIs, ingest endpoints, webhooks	AWS ap-south-1 (Mumbai) — India data residency

3. Data model (event-first)

3.1 Core tables (unchanged from v2.0)

Table	Purpose
users	Identity (phone, email, DPDP status)
consents	Consent per purpose (research, clinician, marketing, on-device-only) with audit trail
events	All time-series events — TimescaleDB hypertable, partitioned by user_id
biomarkers	Structured markers (LOINC-coded value + unit + source + timestamp)
life_events	Ongoing conditions and medical timeline (ICD-10 coded)
therapy_sessions	Halo + external therapy log with parameters
lab_reports	Uploaded PDFs with parsed markers
imaging_studies	MRI / DEXA / X-ray with radiologist notes
clinician_assignments	User ↔ clinician mapping with roles
studies	N-of-1 study definitions
study_arms	Study phases (baseline / on / off / on)
insights	AI-surfaced patterns awaiting clinician review
sinc_conversations	Chat threads with Sinc

3.2 Web-specific tables (new in v3.0)

Table	Purpose
waitlist_signups	Email + source + UTM + timestamp — for launch marketing analysis
press_kit_requests	Journalist name, publication, requested assets, timestamp
investor_data_room_access	Investor email, granted-by, granted-on, revoked-on — for gated /for-investors page
clinician_demo_requests	From /for-clinicians — routed to CMO's queue

3.3 Retention, deletion, audit (unchanged)

- Default retention: user-configurable (30d / 1yr / indefinite); default indefinite for member data, 90d for guest
- Delete request: soft-delete immediately, hard-delete on cron after 7d grace
- Audit log: every read/write of PII to append-only audit_log (immutable, 7-year retention)
- Anonymised research copy: only from users with explicit research consent, separate anon_user_id

4. AI & ML stack — the Sinc engine

4.1 Six layers (updated)

Layer	Function	Technique
L1 Normalise	Cross-device schema, unit harmonisation, outlier flag	Python + pydantic; rule-based
L2 Baseline	Personal baselines per marker, drift detection	Rolling median + robust z-score; per-user model
L3 Pattern surfacing	Event → biomarker correlations with confidence	Within-subject repeated-measures + Bayesian confidence
L4 N-of-1 engine	Structured crossover, washout, honest readout	Multilevel model; pre-registered protocol; "no signal" valid output
L5 Sinc companion (LLM)	Chat, explanations, question suggestion, drafts for clinician review	Frontier LLM (Claude / GPT) with strict RAG on user's own Vault only
L6 Population models	Longevity age, cohort insights, research	Aggregated anonymised data + published biomarker ageing models (Levine, GrimAge, DunedinPACE)

> **Sinc drafts. The clinician signs. This distinction is architecturally enforced.** *Every insight, recommendation, or medical-adjacent statement Sinc generates lands in a clinician-review queue before it is presented to the user as a recommendation. The user sees Sinc's raw observations directly; they never see Sinc's medical suggestions until a clinician has approved them. This is why the sinc-service publishes to a queue consumed by clinician-service, rather than pushing directly to the user's app.*

4.2 System-prompt constraints for Sinc (updated for v3.0)

- You are Sinc, a personal biology companion. You are not a doctor and never diagnose.
- You always cite the specific event or biomarker your observation is based on.
- You NEVER say "this causes" — use "this correlates with" or "this pattern shows."
- Safety-list symptoms (chest pain, suicidal ideation, severe shortness of breath, stroke signs, severe allergic reaction) trigger the emergency protocol and escalate to the assigned clinician immediately.
- Requests outside your competence (drug dosing, diagnosis, prescription) defer to the clinician.
- You surface confidence as a first-class part of every insight ("emerging — 3 events", "building — 6 events", "strong — 12+ events").
- For medical-adjacent recommendations, you draft; the clinician reviews and signs off before the user sees it framed as a recommendation.
- You reference telomeres only in the "telomere-inspired" framing — never as a direct measurement claim unless a telomere-length test product ships.
- You speak in second person, present tense. Names, not roles. Never sycophantic. Never uses "amazing", "great", "awesome."

4.3 The voice pipeline (unchanged from v2.0)

Stage	On device / cloud	Tech
Wake / capture	Device	iOS Core Audio / Android AudioRecord; band BLE audio

Speech-to-text	Device (default)	WhisperKit (iOS) / whisper.cpp (Android) — no cloud
Intent parsing	Device	Small on-device LLM (Phi-3-mini / Gemma-2B) for event tagging
Voice-tonality stress detection	Device	openSMILE feature extraction + local model on prosody
Recap generation	Cloud (opt-in, anonymised)	LLM call; user previews before save
Storage of raw audio	Never by default	Only if user explicitly saves a session for Recap

4.4 MLOps

- Model registry: MLflow (population models); Core ML / TFLite for on-device
- Staged rollout: internal → 5% → 25% → 100%; kill-switch on quality regression
- Every user-facing recommendation carries model version + input hash for audit
- Retraining cadence: population priors monthly; personal models nightly (on-device); no live-user-data cloud training without explicit research consent

5. Mobile app engineering

5.1 Stack (updated)

Layer	Tech	Notes
Framework	React Native + Expo (managed with dev-builds)	iOS + Android from one codebase
Language	TypeScript strict	No any; enforced via ESLint
State	Zustand + TanStack Query	Simple stores; server-state via TanStack
Storage	MMKV (fast KV) + WatermelonDB (SQLite ORM)	Vault local cache + offline queue
Navigation	React Navigation v7	Native stack + bottom tabs
UI system	Design tokens from UI Design System v1.0, styled via Tamagui or NativeWind	Tokens shared with web via a design-tokens package
Charts	Victory Native XL + custom Skia components for the dial + timeline	Matches web chart style — see UI Design System §7
Wearable integrations	react-native-health (iOS), react-native-health-connect (Android), Terra SDK	Terra covers Oura/Whoop/Garmin/Dexcom in one
Voice STT	WhisperKit (iOS), whisper.cpp via native module (Android)	On-device only by default
Sinc chat	Streamed via SSE from sinc-service	Vercel AI SDK or custom
Testing	Jest + Detox (e2e) + Maestro (visual)	CI-gated
Build & deploy	EAS Build + EAS Submit; OTA via EAS Update	Standard Expo workflow

5.2 Performance targets (unchanged)

- App cold start: <2.0s on iPhone 12 / Pixel 6
- Time-to-first-aha on Day 1: <3s after data sync
- Voice log tap-to-transcript: <400ms (on-device)
- Sinc chat first-token latency: <900ms
- 60fps on all animations; use react-native-reanimated

6. Web engineering (NEW in v3.0)

Telomy runs two web applications: telomy.com (marketing) and Telomy Care (clinician dashboard). Both use Next.js 15 with App Router, share the design token package with the mobile app, and deploy to Vercel.

6.1 telomy.com — the marketing site

Full specification is in Website Development Guide v1.0. This section covers only the engineering side.

Layer	Tech	Notes
Framework	Next.js 15 (App Router) + React 19	SSR + ISR for the journal, static for marketing
Language	TypeScript strict	Non-negotiable
Styling	Tailwind CSS 4 + design tokens (shared package)	No CSS-in-JS runtime overhead
Content	Sanity CMS for /journal and /changelog; MDX for /manifesto and /founder	Non-technical team edits copy in Sanity Studio
Animation	Motion One (smaller bundle than Framer Motion)	Only for the hero mark and scroll-in fades
Charts (science page)	Observable Plot with the Telomy chart style	Same style tokens as the mobile app
Icons	Custom SVG set (in-repo, from UI Design System §4)	No Heroicons / Lucide
Forms / waitlist	React Hook Form + Zod + Loops (ESP) via a Next.js Route Handler	Server-side ESP token
Analytics	Plausible (self-hosted or plausible.io)	DPDP-friendly; no cookie banner needed for basic analytics
Hosting	Vercel Pro	Edge network with Mumbai POP; ISR caching
Media / CDN	Cloudinary for photography (AVIF/WebP delivery)	Automatic responsive sizing

6.2 Telomy Care — the clinician web dashboard

Layer	Tech	Notes
Framework	Next.js 15 (App Router)	Separate Vercel project; app.telomy.com/care route
Auth	Same identity-service as mobile app; SSO via magic link + TOTP	Clinician role required
State	TanStack Query + Zustand for local UI state	Same patterns as mobile
Data-viz	Observable Plot + custom Telomy chart theme	Reused across surfaces
UI system	Design tokens from UI Design System v1.0; shadcn/ui as component base, restyled with Telomy tokens	Faster to build than from scratch; still Telomy-branded
Real-time	SSE for new insights arriving in queue; websockets for multi-user editing (later)	Standard patterns
Print / export	Server-side PDF via Puppeteer for clinician reports	For patient handouts, insurance filings

6.3 The shared design-tokens package

A single npm package @telomy/design-tokens exports the token system defined in UI Design System v1.0 in three formats:

- CSS custom properties for web (imported once at the root)
- A JavaScript object for React Native styling
- A Figma Tokens JSON for the design team

The package is published from a design-tokens repo. Changes go through PR review. When a token changes, both apps update on next build.

6.4 Cross-surface auth

Scenario	Behaviour
User signs in on mobile	Auth token cached; visits telomy.com in same browser sees personalised bar
Clinician signs in on Telomy Care	Same identity-service; role check restricts to clinician endpoints
User opens Telomy Care preview link from mobile	Deep link with short-lived token; opens patient's own Vault view
Consent revoked on any surface	Propagates to all surfaces within 5 minutes; background job invalidates cached views

6.5 Web performance targets

- Lighthouse Performance: ≥95 mobile, ≥98 desktop, at launch
- LCP: ≤1.5s on mobile 4G, ≤800ms on desktop broadband
- CLS: <0.05 across all pages
- Total JavaScript bundle: ≤150 KB gzipped on homepage; ≤100 KB on content pages
- Total page weight: ≤1.5 MB on homepage including hero imagery
- Self-hosted fonts with font-display: swap and subsetting

6.6 AI-readability (new consideration)

Both web apps are structured for LLM ingestion. See Website Development Guide §20.5 for the full spec: clean semantic HTML, JSON-LD FAQPage schema on homepage, /llms.txt file at root, canonical URLs, and every claim citation-linked.

7. Backend services & APIs

7.1 Eight FastAPI microservices

Service	Owns	Key endpoints
vault-service	Vault reads/writes, event storage	POST /events, GET /events, GET /biomarkers
ingest-service	Wearable webhooks, lab PDF upload, imaging upload	POST /ingest/{oura, apple, lab, imaging}
insight-service	Pattern detection, correlation compute, longevity age	GET /insights, POST /correlations/compute
study-service	N-of-1 study lifecycle	POST /studies, GET /studies/{id}, POST /studies/{id}/finalize
sinc-service	LLM chat, RAG orchestration, tool execution, insight drafting	POST /sinc/chat (SSE), POST /sinc/recap, POST /sinc/draft-insight
clinician-service	Telomy Care queue, sign-off, alerts, protocol authoring	GET /clinician/queue, POST /clinician/sign-off, POST /clinician/protocols
identity-service	Auth, DPDP consent, DSR portal, cross-surface SSO	POST /auth/otp, POST /consent, GET /export, DELETE /account
payments-service	Razorpay, subscriptions, Telomy Formulas marketplace	POST /orders, POST /subscriptions, POST /webhooks/razorpay

7.2 API conventions

- REST + JSON for CRUD; Server-Sent Events for streaming (Sinc chat); gRPC internally between services
- OpenAPI 3.1 spec per service; auto-generated TypeScript client for mobile app and web apps
- Idempotency-Key header required on all POST /events and /orders
- Rate limits at gateway: 100 req/min/user default; 10 req/min for Sinc chat
- Auth: JWT (30 min) + refresh token (30 days), rotated on use
- Every request carries x-purpose header (self/research/clinician) enforced against user consent

8. Third-party integrations

Integration	What	How
Terra API (or Rook, Vital)	Wearables aggregator — Oura/Whoop/Garmin/Dexcom/Abbott	Webhook subscription; ~$0.50/user/month
Apple HealthKit	iOS wearable data	react-native-health; user-side OAuth
Google Health Connect	Android wearable data	react-native-health-connect
Redcliffe / Healthians / Tata 1mg	Home lab collection	Partner API (private) + booking widget
Razorpay	Payments, subscriptions	Standard SDK + webhooks
Twilio / Exotel	OTP auth	SMS gateway
Sanity CMS	Website content (journal, changelog, pages)	GROQ queries from Next.js server components
Cloudinary	Photography and product imagery for web	Direct upload + on-the-fly transformations
Plausible	Web analytics (privacy-first)	Script include; no cookie required
Loops	Waitlist ESP + broadcast emails	Server-side API from Next.js route handler
Whisper API (fallback)	STT if on-device fails	Rare path; anonymised text only
Anthropic Claude / OpenAI GPT	Sinc LLM	API with strict prompt guardrails; no PHI in prompt fingerprints

9. Security & DPDP compliance

> **India's DPDP Rules operational since November 2025. Penalties up to ₹250 crore. Health data is high-risk. Every architectural decision below is a compliance decision, not just a security one.**

9.1 Control matrix

Control	Implementation
Encryption at rest	AES-256 (AWS KMS); customer-managed keys for enterprise clinics
Encryption in transit	TLS 1.3+ everywhere; mTLS between services
Key management	AWS KMS with rotation every 90 days; keys never leave HSM
Consent per purpose	Every API call carries x-purpose; consent service is authoritative gate
Data minimisation	PHI never in URLs, logs, error messages, prompt-model fingerprints
Audit log	Immutable append-only log to S3 Object Lock; 7-year retention
Data residency	India-first (ap-south-1); export markets use local regions
Breach notification	Automated pager to DPO on any anomaly; 72h clock; template statements ready
Data-subject rights	In-app portal on mobile + web: access, correction, deletion; SLAs enforced via cron
Consent Manager (ABDM)	Integrated by Nov 2026 per DPDP Rules
Vulnerability program	Annual pen test (CERT-In empanelled firm) + private bug bounty
Insider access	Zero standing access to prod data; every access via time-boxed break-glass with auto-audit
Web CSP	Strict Content-Security-Policy on telomy.com and Telomy Care; no inline scripts
Web session security	HTTPS-only cookies; SameSite=Lax; short JWT expiry

9.2 AI/ML safety

- **Hallucination in medical context** — RAG-only responses; sanitiser strips uncited claims; clinician sign-off for recommendations
- **Bias from population models** — explicit population labels ("for South Asian males"); model cards; clinician review for edge cases
- **Confidence over-claim** — every insight surfaces evidence count + CI; "no reliable signal" is a valid output
- **Prompt injection via user notes** — user text is quoted, not interpreted, in system prompts; tool calls sandboxed
- **Web LLM crawler abuse** — rate-limit LLM crawlers on /journal to prevent scraping abuse; welcome citation crawlers per robots.txt

10. DevOps, environments & observability

10.1 Environments

Env	Purpose	Data
local	Developer laptops	Synthetic seed data
ci	GitHub Actions runs	Ephemeral
dev	Shared dev	Synthetic + a few real staff accounts
staging	Pre-prod, mirrors prod	Anonymised prod subset (with consent)
prod	Live	Real user data, DPDP-controlled
preview (web)	Vercel per-PR previews for telomy.com + Telomy Care	Synthetic data; no PHI

10.2 CI/CD

- GitHub Actions: lint → typecheck → unit → integration → build → deploy
- Every PR runs the full test suite + Snyk security scan
- Backend deploys via GitOps (ArgoCD) with automatic rollback on error-rate spike
- Web deploys via Vercel (auto per push to main); preview URLs per PR for stakeholder review
- Mobile: EAS Build for iOS/Android; OTA JS updates via EAS Update
- Database migrations: alembic, run pre-deploy with lock-safety checks
- Design tokens: separate repo with semver; consumer apps update via dependabot

10.3 Observability

Layer	Tool	Purpose
Logs	Grafana Loki (backend), Vercel Logs (web)	Structured JSON, PHI-redacted, 90d retention
Metrics	Prometheus + Grafana	Golden signals per service
Traces	OpenTelemetry → Grafana Tempo	End-to-end request tracing
Errors	Sentry (backend + mobile + web)	Aggregated stack traces (PHI-scrubbed)
Alerts	Grafana OnCall / PagerDuty	Tiered severity; DPO on data alerts
Product analytics	Mixpanel or Posthog (self-hosted, consent-gated)	Opt-in for research consent only
Web analytics	Plausible (privacy-first)	DPDP-friendly; no cookie banner
Real user monitoring	Vercel Speed Insights on web	Core Web Vitals tracking

10.4 SLOs & DR

- API availability: 99.9% (43min/month downtime budget)
- Vault write latency (p99): <150ms
- Sinc chat first-token (p95): <900ms

- Web LCP (p95): <1.5s on mobile 4G
- RPO 15min / RTO 1h; nightly encrypted backups to a separate region
- Runbook + game-day exercises quarterly

11. Design system integration (NEW in v3.0)

The UI Design System v1.0 is the canonical source for how Telomy looks and behaves. This section explains how the design system integrates into the engineering codebase across the three surfaces.

11.1 The design-tokens package (@telomy/design-tokens)

Published as a private npm package from a design-tokens repository. Contains:

- colours.ts — every colour token as a TypeScript constant, plus a CSS custom-property file
- typography.ts — font families, weights, sizes, line-heights as constants + Tailwind theme config
- spacing.ts — the 8pt system as constants + Tailwind theme config
- motion.ts — durations and easing curves as constants + Framer/Motion One presets
- figma-tokens.json — the source of truth for Figma variable sync

Updates flow: designer changes token in Figma → design lead PRs the JSON export to the design-tokens repo → PR merged, package bumps → dependabot PRs open in mobile app, telomy.com, Telomy Care repos → developers merge those PRs → next build picks up new tokens.

11.2 Component library — @telomy/ui

A second package containing the components defined in UI Design System §6–§9, published in two variants:

- Web (React) — for telomy.com and Telomy Care
- Native (React Native) — for the mobile app

Both variants share prop APIs and behaviour; only the rendering primitives differ. This means the mobile app and Telomy Care can share screen-level code where it makes sense (e.g. the consent flow is the same UX everywhere).

11.3 Cross-surface consistency enforcement

- A shared Storybook (web) and React Native equivalent (mobile) — both consuming the same design-tokens package.
- Visual regression tests on every PR via Chromatic (web) and Maestro (mobile).
- Design review checklist embedded in the PR template — the last three items are always token compliance, accessibility, and cross-surface consistency.
- Monthly design-system sync — designer + developer walk through drift; any drift is either resolved or documented as an accepted variance.

12. Engineering team & 6-month build plan

12.1 Team (updated in v3.0)

Role	Count	When
Technical co-founder / CTO	1	Pre-seed gate — URGENT
Mobile lead (React Native)	1	Month 1
Backend / data lead (Python)	1	Month 1
Frontend web lead (Next.js) — NEW in v3.0	1	Month 1 (also owns Telomy Care)
Founding mobile engineer	1	Month 2
Founding backend engineer	1	Month 2
ML / signal engineer	1	Month 3
Cloud / DevOps	1 (fractional → full-time Month 4)	Month 2
Clinician-side product designer + web engineer for Telomy Care	0.5 designer + 1 engineer	Month 4 (web engineer bumped up)
QA / test engineer	1	Month 4
Founding designer (brand + product)	1	Month 1 (works with brand designer from Brand Identity Brief)

12.2 6-month build plan (updated)

Month	Deliverable
1	Repo, CI/CD, environments, design-tokens package published, auth service, event schema locked, telomy.com scaffold live at /soon
2	Telomy Vault MVP: mobile app + Apple Health + event ingest + on-device storage; telomy.com Journal + Manifesto live
3	Voice log + on-device Whisper + adaptive scroll menu; Sinc chat MVP (RAG on Vault); telomy.com launch
4	Lab PDF upload + RAG parsing; Telomy Care web dashboard MVP; first design-partner clinic pilot
5	Pattern surfacing (L3) + first insights on Home; longevity age v1; DPDP consent manager
6	N-of-1 study engine v1; Telomy Formulas marketplace + Razorpay; 50-user closed pilot; seed round

13. Closing & open decisions for the incoming CTO

Decisions the CTO should re-evaluate at Month 1 with fresh eyes:

- React Native vs Flutter — this doc recommends RN (larger India talent pool, better web-code-share via Expo Router); both are defensible.
- Cloud provider — AWS (Mumbai) vs GCP (Mumbai). Both meet DPDP. AWS has more India talent; GCP has better ML tooling. Depends on ML priorities.
- LLM vendor for Sinc — Claude vs GPT vs open-source (Llama 3.3, Mistral). Depends on cost, latency, DPA terms. Ideally start with Claude for the honesty properties, benchmark alternatives quarterly.
- Terra vs Rook vs Vital for the wearable aggregator — depends on Y1 integration list.
- In-house vs partner for lab PDF parsing — partner recommended for Y1 (Roboflow document AI or similar), in-house at Series A.
- Sanity vs Contentful vs Payload for the CMS — Sanity recommended (developer experience, cost); Payload if self-hosting is a hard requirement for editorial content.
- shadcn/ui as component base for Telomy Care — recommended (fast to build, easy to restyle); alternatives are Radix + custom or Ariakit.
- Design system as monorepo vs multi-repo — recommended monorepo (Turborepo) if the same team owns app + web; separate repos if they're distinct teams.

The design system + tokens strategy is settled and should NOT be re-evaluated — changing it after Month 2 costs weeks of rework across surfaces. The tokens repo is the single source of truth; the UI Design System v1.0 is its documentation.

Where human biology becomes intelligent.

— End of Engineering Guide v3.0 —

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